Breaking the Buzz Words of AI Field

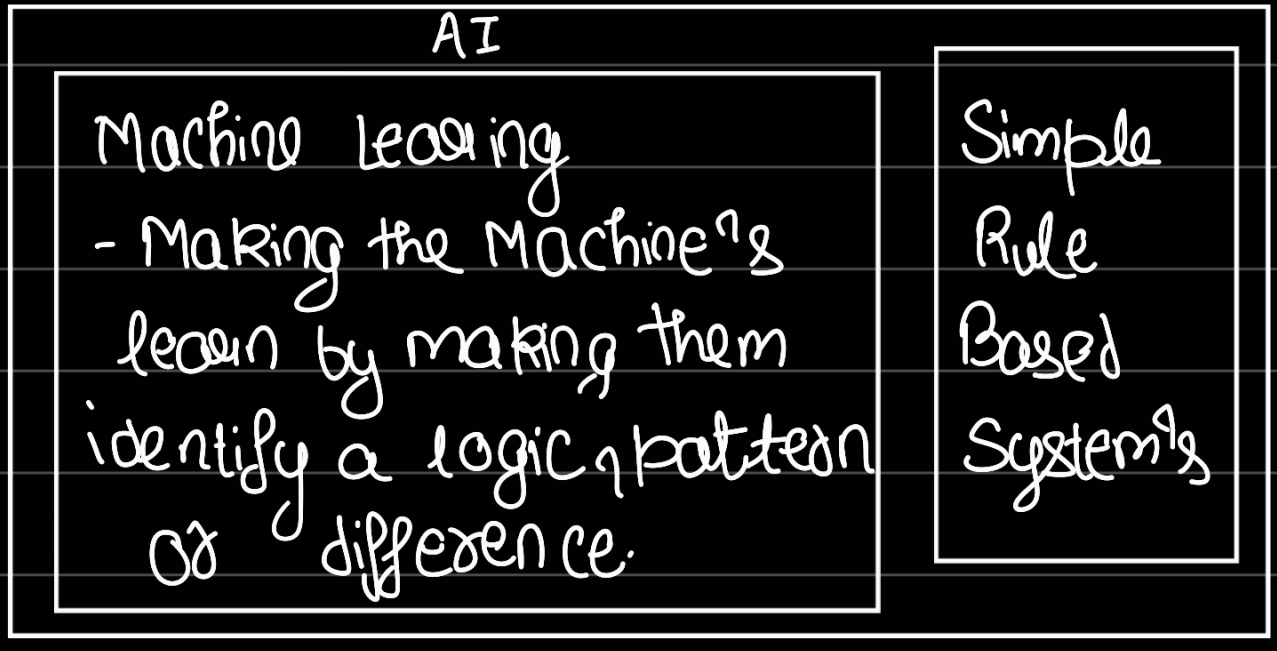
- Sayed Zahuel

AI: Simulating any tasks that needs Human or Human-like Intelligence through any program is called Artificial Intelligence

We have lots of words in this Industry like

Machine Learning
Deep Learning
Generative AI
Data Science

We will understand these step by step & know their hierarchy using Venn Diagrams



While Developing AI System's we use ML when

- We need the system to learn from data or improve over time

- The rules are too complex to write manually

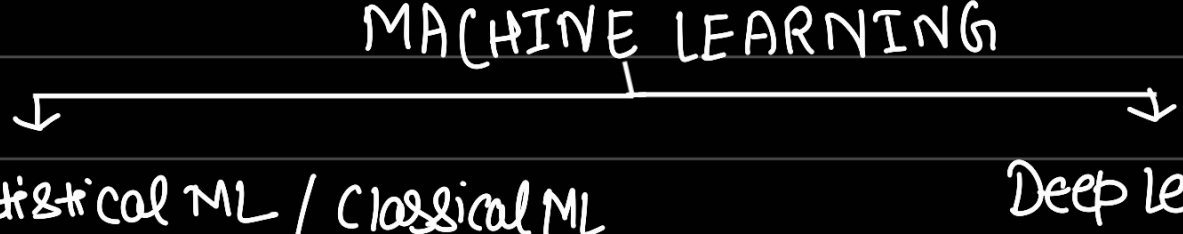
- Each input can have different cases

and we use Simple - rule-based Systems when the logic is clear & doesn't change much

Example: Traffic control system's that work on simple logics like

- If 30 seconds passed at current light make it make the adjacent light green

Now, As we covered till here. We will now cover ML in detail:



Statistical / Classical ML

- We use this when we have one type of Structured Data

In this method, we have techniques to infer pattern from the Data like

- Linear Regression : Predict house prices based on size, no. of rooms etc

- Decision tree / Random Forest : Classify whether a customer will buy or not

But, The Big Guy Deep Learning enters the space when

- When we have a large amount of unstructured Data with more than one datatype (images, audio, text)

Deep learning works on a mechanism of neural networks that is basically a computer model inspired by human brain that learn patterns from data by adjusting connections between layers of simple units called neurons

- So Basically Deep learning learns patterns automatically without needing manual feature design

Lets understand a case of Stat ML vs Deep learn

Case: whether the image is a dog

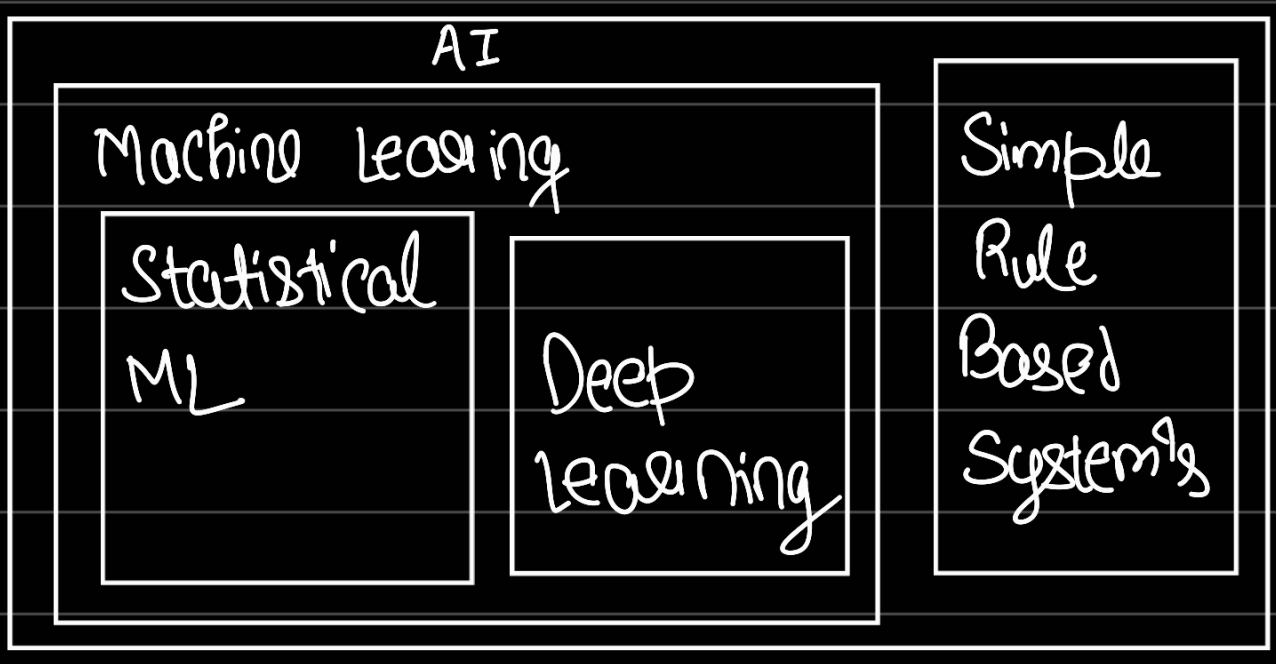
Stat ML	Deep learning
You need to describe the image before giving it on things like - No. of brown pixel - No of legs, Tail	Here you just need to gives a laarge amount of dog photos

- Design of claws

Using these all it will create a model & gives the ans on that calculation

By studying those photos the model figures out that the image is dog or not all by itself

So, Now the Ven diagram becomes

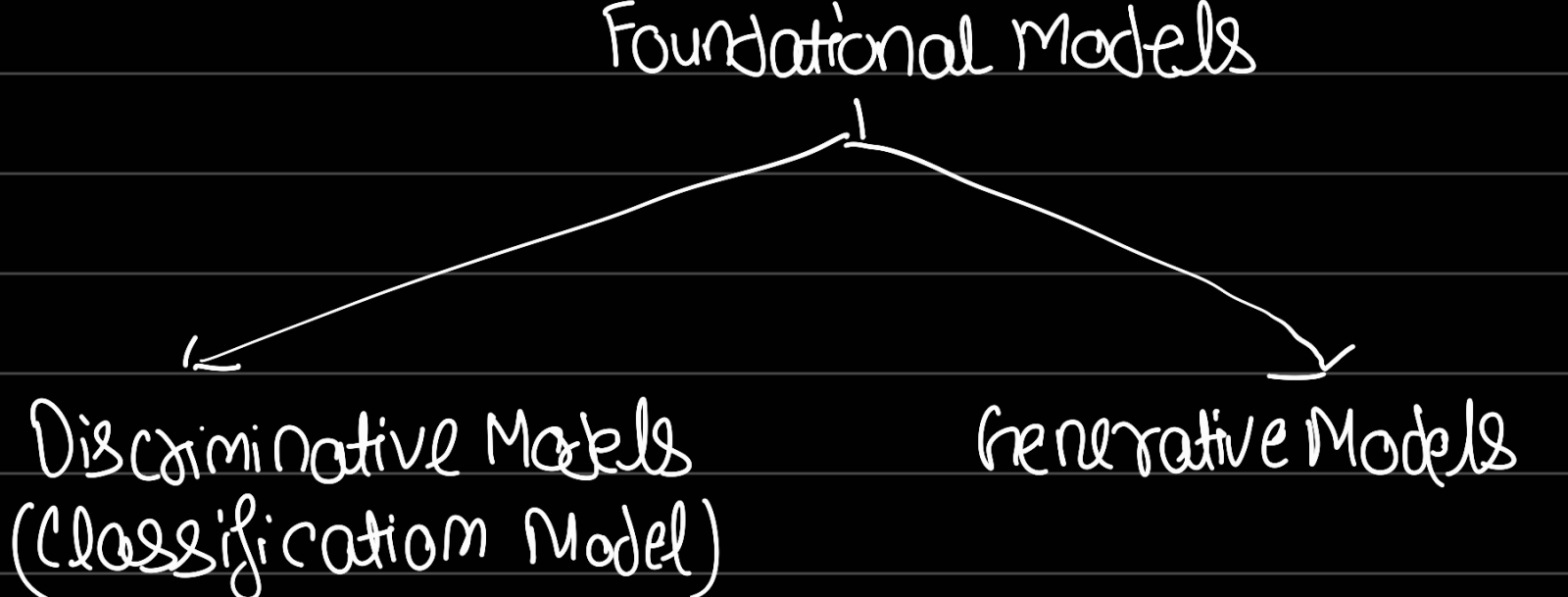


In Deep Learning we have a lot of Neural Networks like
- ANNs
- CNNs
- RNNs
- Transformers

Using Transformers & the AI would got some crazy thing called as

Foundations Models

These were large AI models trained on massive amounts of Diverse data, which can be adapted for many different tasks



- Classification Models predict labels of classify inputs into categories

Example: Spam or Not Spam, Sentiment Analysis

- Generative Models

These models generates new content using user instruction said as prompt acc to user needs

All the LLMs comes under these Generative Models as they are designed to

- generate text
- continue conversations
- summarizes text

We have many types of Generative Models like

Text to Text : GPT 3.5
Text to Image : Dalle
Text to Video : Sora, Veo
Text to Speech : Eleven Labs, Deepgram

All these generative Model's comes under the Gen AI Field inside the Deep Learning Architecture

- Now the Venn Diagram becomes

Now last thing is left that is Data Science

Data Science is basically using all these AI tools with some maths like Probs, Stats and calculus to get some data-driven insights useful for decision making

